

The efficacy and safety of preoperative glucocorticoid in herniorrhaphy: A systematic review and meta-analysis (protocol)

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Short running title: preoperative glucocorticoid in hernia repair

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1. Introduction

Inguinal and abdominal hernia repair surgery is one of the most common surgeries and is performed on more than 20 million people worldwide each year.¹ The duration of convalescence after herniorrhaphy is of great socioeconomic concern, and herniorrhaphy has an approximately 6% prevalence of debilitating pain affecting normal daily activities or works.² Herniorrhaphy is the most painful on postoperative day 0 and less painful on postoperative day 3.³ Postoperative pain is most often treated with non-steroidal anti-inflammatory drugs or opioids, but opioids primarily have the side effects of nausea.⁴ The prevention of pain and emesis after herniorrhaphy is thus crucial due to achieve rapid recovery and return to normal activities.⁵

Glucocorticoids including dexamethasone, betamethasone, and methylprednisolone, have anti-inflammatory, analgesic (opioid-sparing), and antiemetic effects.^{6,7} Long-term treatment with glucocorticoids has many adverse effects, so a single preoperative glucocorticoid administration is recommended.⁸ In previous systematic reviews, preoperative glucocorticoid administration may lead to improved recovery after thyroidectomy,⁹ total hip arthroplasty,¹⁰ tonsillectomy,¹¹ spinal fusion surgery,¹² and breast surgery.¹³ However, the efficacy of preoperative glucocorticoid administration in herniorrhaphy has never been systematically reviewed and analyzed.

Therefore, this study will aim to assess the efficacy and safety of preoperative glucocorticoid administration in herniorrhaphy.

2. Methods

We followed the Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 for preparing this protocol.¹⁴ We will publish this protocol in protocols.io (<https://www.protocols.io/>).

2.1 Criteria for considering studies for this review

2.1.1 Types of studies

We will include randomized individual controlled trials (RCTs). We will include studies reported as full text, those published as abstract only, and unpublished data. We will exclude cross-over trials, cluster trials, and non-randomized trials.

2.1.2 Types of participants

We will include trials comparing adults aged over 15 years who scheduled for herniorrhaphy as the herniorrhaphy over the age of 15 is generally the same as that for adults.¹⁵ We will exclude participants with the following co-morbidities/characteristics: daily use of systemic glucocorticoid and known adverse reactions to glucocorticoid. We will include trials comparing intravenous preoperative glucocorticoid administration with placebo or usual care. We will include the following co-interventions provided they are not part of the randomized treatment: antibiotics, analgesics, and antiemetics.

2.2 Types of outcome measures:

2.2.1 Primary outcomes

1. The mean score of the visual analog scale (VAS)/ numeric rating scale (NRS) for participants' pain at postoperative day 0 after herniorrhaphy regardless of at rest or during activity. If pain has been measured multiple times at postoperative day 0, the last score will be adopted. If pain at rest and during activities are measured, pain at rest will be used.
2. The rate of patients requiring rescue analgesics, defined as the number of patients requiring rescue analgesics divided by total patients' number during postoperative stays.
3. The rate of patients vomiting, defined as the number of patients vomiting divided by total patients' number within postoperative day 7.

2.2.2 Secondary outcomes

1. The mean score of the VAS/ NRS for participants' pain at postoperative day 1 after herniorrhaphy regardless of at rest or during activity. If pain has been measured multiple times at postoperative day 1, the initial score will be adopted. If pain at rest and during activities are measured, pain at rest will be used.
2. Lengths of the postoperative stays
3. The mean score of the visual analog scale (VAS)/ numeric rating scale (NRS) for participants' fatigue at postoperative day 1 after hernia repair surgery. If fatigue has been measured multiple times at postoperative day 1, the initial score will be adopted.
4. All adverse events defined by the author

2.3 Search methods for identification of studies

2.3.1 Electronic searches

We will conduct a literature search to identify all published and unpublished randomized controlled trials in all languages. We will translate non-English language papers and fully assess them for potential inclusion in the review as necessary.

We will search the following electronic databases:

1. Cochrane Central Register of Controlled Trials(CENTRAL) ;
2. MEDLINE via Ovid (1966 to present);
3. EMBASE (1988 to present);

See Appendix 1 for the search strategies.

2.3.2 Searching other resources

We will check the reference lists of all primary studies and review articles for additional references.

We will also conduct a search of clinical trial registers/trial result registers:

1. Clinical Trials.gov

2. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP)

See Appendix 2 for the search strategies.

We will check the references of guidelines. We will contact authors of identified trials and ask them to identify other published and unpublished studies.

2.4 Data collection and analysis

2.4.1 Selection of studies

Two of the three review authors (JW and KR or TK) will independently screen titles and abstracts for inclusion. All of the potential studies we identify as a result of the search will be coded them as either ‘retrieve’ (eligible, potentially eligible, or unclear) or ‘do not retrieve’. We will retrieve the full text of potentially eligible studies and two of the three review authors (JW and KR or TK) will independently screen the full text, and identify studies for inclusion, and identify and record reasons for exclusion of the ineligible studies. We will resolve any disagreement through discussion or, if required, we will consult a third author (KK). We will identify and exclude duplicates and collate multiple reports of the same study, so that each study rather than each report is the unit of interest in the review. We will record the selection process in sufficient detail to complete a PRISMA flow diagram.

2.4.2 Data extraction and management

We will use a standardized data collection form for study characteristics and outcome data, which will be piloted on at least one study included in the review. Two of the three review author (JW and KR or TK) will independently extract study characteristics from included studies. We will extract the following study characteristics: study design, number of study centers and location, types of hernia surgery and anesthesia, date of study, participants number, and mean age.

Two of the three review authors (JW and KR or TK) will independently extract outcome data from the included studies. We will note in the 'Characteristics of included studies' table if the study authors reported outcome data was reported in an unusable way. We will resolve disagreements by consensus or by involving a third review author (KK). We will copy across the data from the data collection form into the Review Manager 4 (RevMan 4.4.1) file.

2.5 Assessment of risk of bias in included studies

Two of the three review author (JW and KR or TK) will independently assess the risk of bias using the Risk of Bias 2.¹⁶ We will resolve any disagreement by discussion, or by involving a third review author (KK).

2.6 Measures of treatment effect

We will analyze dichotomous data of rescue analgesics and vomiting patients as risk ratios, and continuous data of patients' pain, fatigue, and hospital stays as mean difference or standardized mean difference. We will ensure that higher scores for continuous outcomes have the same meaning for the particular outcome, explain the direction to the reader and report where the directions were reversed if this was necessary. We will summarize adverse events based on the definition by the original article, but we will not perform meta-analysis.

2.7 Unit of analysis issues

Multiple comparisons

All intervention groups that are relevant to this review will be included.

Multiple outcomes

If pain has been measured multiple times at postoperative day 0, the last score will be adopted. If pain has been measured multiple times at postoperative day 1, the initial score will be adopted. If

pain at rest and during activities are measured, pain at rest will be used. If fatigue has been measured multiple times at postoperative day 1, the initial score will be adopted.

2.8 Dealing with missing data

We will ask not-presented data to the original authors.

2.8.1 Missing outcomes

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible. For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook.¹⁷ We will perform meta-analysis about the available data in the original study.

2.8.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman.¹⁸ If we don't know these values when we contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook¹⁷, or validated method.¹⁹ Validity of these methods will be analyzed by sensitivity analysis.

2.9 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I^2 statistic (I^2 values of 0% to 40%: might not be important; 30%to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity).¹⁷ When there is substantial heterogeneity ($I^2 > 50\%$), we will assess the reason of the heterogeneity. Cochrane Chi²test (Q-test) will be performed for I^2 statistic, and P value less than 0.10 will be defined as statistically significant.

2.10 Dealing with reporting data

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. If we are able to pool more than 10 trials, we will create and examine a funnel plot to explore possible publication biases.¹⁷ We will use Egger's test to determine the statistical significance of the reporting bias. We will consider $P < 0.05$ to be a statistically significant reporting bias.

2.11 Data synthesis

We will perform the meta-analysis using Review Manager software (RevMan 5.4.1). We will use random-effects model by default.

2.12 Subgroup analysis and investigation of heterogeneity

We plan to carry out the following subgroup analyses:

1. Hernia types (inguinal vs. abdominal hernia)
2. Surgical approach (open vs. laparoscopy)
3. Steroid types (dexamethasone vs. betamethasone vs. triamcinolone vs. prednisolone vs. fluorometholone)

2.13 Sensitivity analysis

We will perform sensitivity analysis defined to assess the robustness of our conclusions. This will involve:

1. Exclusion of studies using imputed statistics.
2. Only the participants who complete the study with complete data

2.14 'Summary of findings' table

Summary of findings table will be made for the following outcome based on the Cochrane handbook.¹⁷ We will include grading to evaluate the quality of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach for each Summary of findings table.²⁰

3. Conflict of Interest

The authors declare no conflicts of interests.

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Appendix 1: Search strategy of the electronic databases

CENTRAL search strategy

#1. MeSH descriptor: [Hernia, Abdominal] explode all trees

#2. MeSH descriptor: [Hernia, Umbilical] explode all trees

#3. MeSH descriptor: [Hernia, Inguinal] explode all trees

#4. MeSH descriptor: [Hernia, Femoral] explode all trees

#5. #1 OR #2 OR #3 OR #4

#6. (Inguina* OR temporal OR groin* OR abdominal OR umbilical OR femoral):ti,ab,kw (Word variations have been searched)

#7. herni*:ti,ab,kw (Word variations have been searched)

#8. #6 AND #7

#9. #5 OR #8

#10. MeSH descriptor: [Adrenal Cortex Hormones] explode all trees

#11. (steroid* OR glucocorticoid* OR dexamethasone* OR betamethasone* OR triamcinolone* OR prednisolone* OR fluorometholone*):ti,ab,kw (Word variations have been searched)

#12. #10 OR #11

#13. #9 AND #12

MEDLINE (via PubMed) search strategy

#1. "Hernia, Abdominal"[Mesh] OR "Hernia, Umbilical"[Mesh] OR "Hernia, Inguinal"[Mesh] OR "Hernia, Femoral"[Mesh]

#2. (Inguina*[tiab] OR temporal[tiab] OR groin*[tiab] OR abdominal[tiab] OR umbilical[tiab] OR femoral[tiab]) AND herni*[tiab]

#3. #1 OR #2

#4. "Adrenal Cortex Hormones"[MeSH]

#5. steroid*[tiab] OR glucocorticoid*[tiab] OR dexamethasone*[tiab] OR betamethasone*[tiab] OR triamcinolone*[tiab] OR prednisolone*[tiab] OR fluorometholone*[tiab]

#6. #4 OR #5

#7. #3 AND #6

#8. (randomized controlled trial [pt] OR controlled clinical trial [pt] OR randomized [tiab] OR drug therapy[sh] OR placebo [tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab]) NOT (animals [mh] NOT humans [mh])

#9. #7 AND #8

EMBASE (via ProQuest Dialog) search strategy

S1 EMB.EXACT.EXPLODE("abdominal wall hernia")

S2 EMB.EXACT.EXPLODE("umbilical hernia")

S3 EMB.EXACT.EXPLODE("inguinal hernia")

S4 EMB.EXACT.EXPLODE("femoral hernia")

S5 ab(Inguinal OR temporal OR groin OR abdominal OR umbilical OR femoral) OR ti(Inguinal OR temporal OR groin OR abdominal OR umbilical OR femoral)

S6 ab(hernia) OR ti(hernia)

S7 S5 AND S6

S8 S1 OR S2 OR S3 OR S4 OR S7

S9 EMB.EXACT.EXPLODE("corticosteroid")

S10 ab(steroid OR glucocorticoid OR dexamethasone OR betamethasone OR triamcinolone OR prednisolone OR fluorometholone) OR ti(steroid OR glucocorticoid OR dexamethasone OR betamethasone OR triamcinolone OR prednisolone OR fluorometholone)

S11 S9 OR S10

S12 S7 AND S11

S13 (ab(random*) OR ti(random*)) OR (ab(placebo*) OR ti(placebo*)) OR (ab(double NEAR/1 blind*) OR ti(double NEAR/1 blind*))

S14 S12 AND S13

Appendix 2: Clinical trial registries

ClinicalTrials.gov. search strategy

Condition or disease: ((Inguinal OR temporal OR groin OR abdominal OR umbilical OR femoral) AND hernia)

Intervention/treatment: (steroid OR glucocorticoid OR dexamethasone OR betamethasone OR triamcinolone OR prednisolone OR fluorometholone)

ICTRP search strategy

((Inguinal OR temporal OR groin OR abdominal OR umbilical OR femoral) AND hernia) AND (steroid OR glucocorticoid OR dexamethasone OR betamethasone OR triamcinolone OR prednisolone OR fluorometholone)